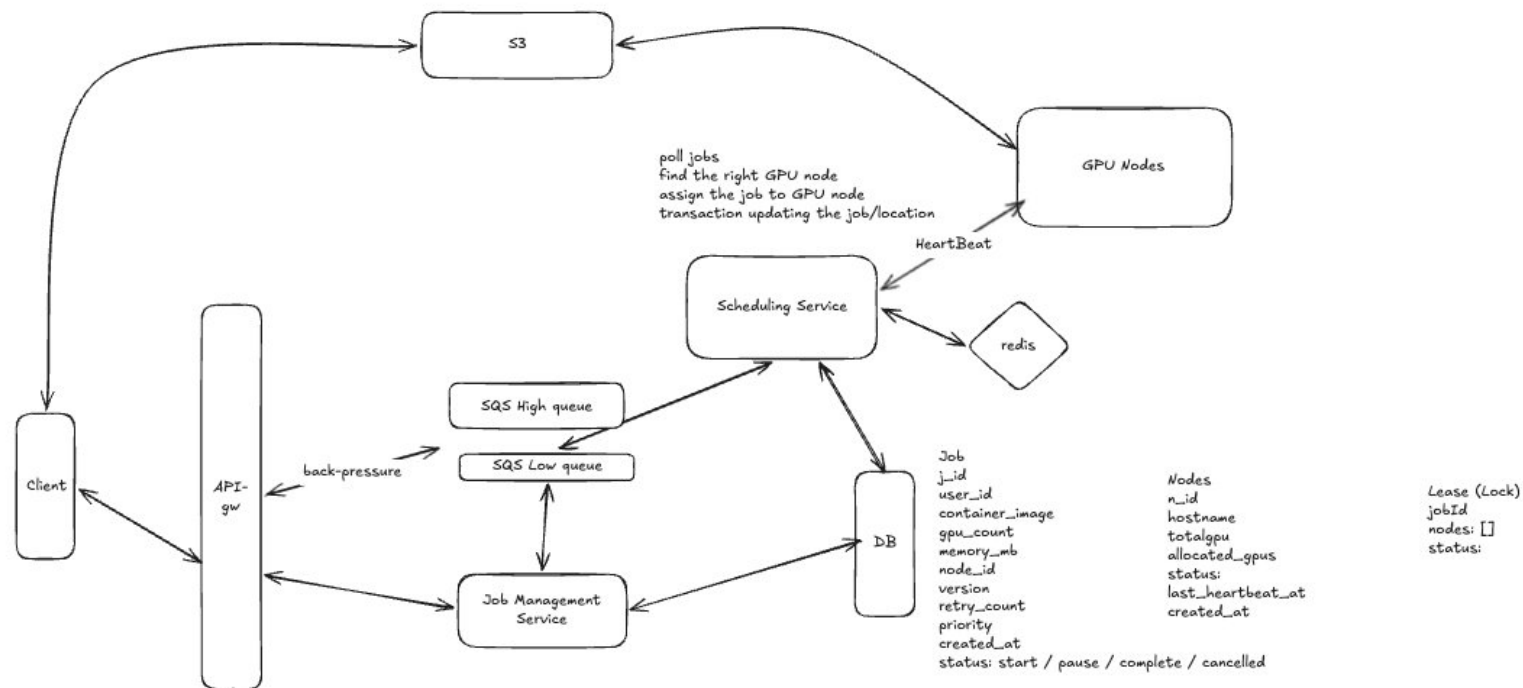


Functional requirements:

1. Submit job with GPU / resource specification;
2. Assign the jobs to available GPU nodes based on priority / resource fit;
3. User can query job status
4. Preempts lower-priority jobs when a higher-priority job arrives and no resources;

Non-functional requirement:

1. Job consistency: (double assign a job, double assign GPU, ghost GPU)
2. Scalability: 1000 GPU Nodes, 10K queued jobs, 100 concurrent jobs
4. No Starvation:
4. Fault-tolerance: auto-recover the jobs;



STRIC ACID trnasactino guarantee on the database updates, to prevent (assign two jobs in one CPU)

lease to prevent ghost GPUs

heartbeat to sync data with db + FENDING_TOKEN + Optimistic lock

to prevent stale updates

agent killing the jobs to prevent job double assign

Strictly follow state transition sequence

reconcile scan jobs to release resources;